W5.1 Classification accuracy and cross validation

Tuesday, October 6, 2026 8:53 AM

2 goals: ① Wrap ch 4, classification accuracy.
② Ch5 intro (resampling for cross validation & bootstraps + CI + HT)
Ch3

Ch 4 Classification $Y = f(X) + \varepsilon$, $Y \in \{Y_1, Y_2, \dots, Y_c\}$
$c \geq 2$

How to assess accuracy?

Ex $Y \in \{0, 1\} = \{\text{ham}, \text{spam}\}$ (focus on spam det)
pos. good email bad email

$\hat{Y} \approx Y$
model pred true label

Confusion matrix

Model preds	True labels: 0 = Ham (TNR)	1 = Spam (TPR)	
0	93 TN	1 FN	94
1	2 FP	4 TP	6
	95	5	n = 100

TN = true negatives FN = false negative
TP = true positives FP = false positive

(M1) Accuracy: $Acc = \frac{TN + TP}{n} = \frac{93+4}{100} = 0.97$ ← misleading WRT class imbalance ← many metrics.

Base line model Always predict $\hat{y} = 0$ 95% acc.

AKA sensitivity or recall.

(M2) True positive rate: $TPR = \frac{TP}{TP+FN} = \frac{4}{5} = \boxed{0.8}$ ← flag as many 1 correctly, ignores incorrect 1's

Specificity

(M3) True negative rate: $TNR = \frac{TN}{TN+FP} = \frac{93}{95} = 0.979$ ← Same as TPR but for 0's.

(M4) Precision: $Prec = \frac{TP}{TP+FP} = \frac{4}{6} = \boxed{0.\bar{6}}$ ← % flagged as 1 are correct.

(M5) F_1-score: $F_1 = \left[2 \frac{Prec \cdot Recall}{Prec + Recall} \right] = 2 \frac{\frac{4}{6} \cdot \frac{4}{5}}{\frac{4}{6} + \frac{4}{5}} = 2 \frac{\frac{16}{30}}{\frac{20+24}{30}}$

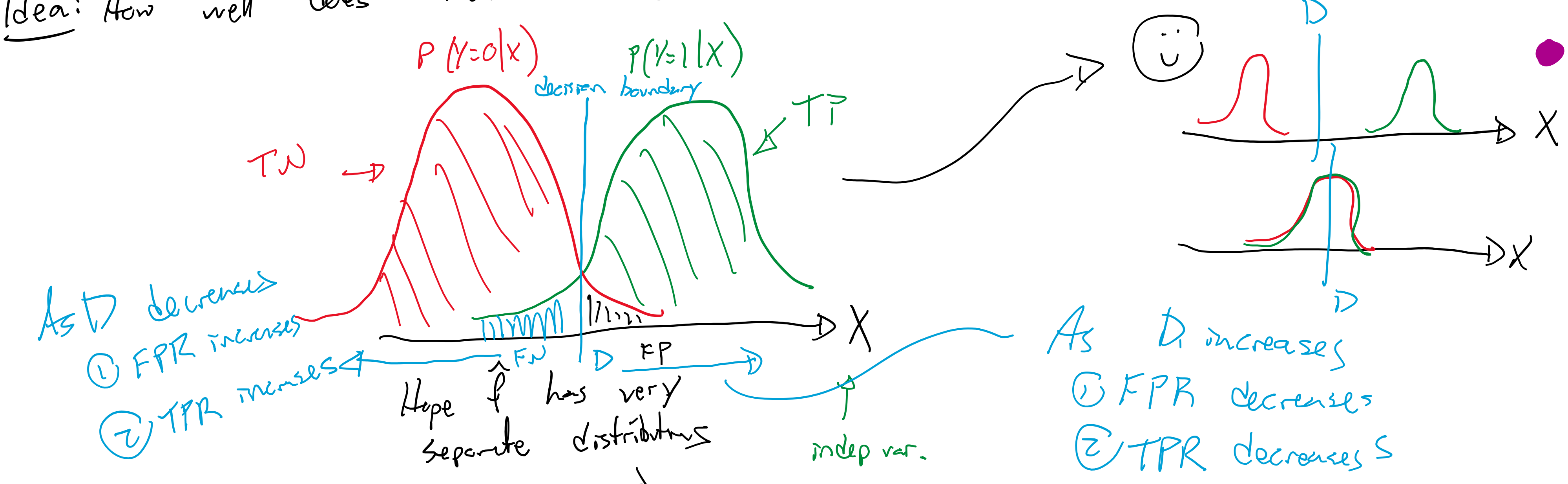
$= \cancel{2} \cdot \frac{16}{30} \cdot \frac{30}{44} = \frac{16}{22} = \frac{8}{11} = 0.727$ in $[0.\bar{6}, 0.8]$ skews to smallest.
prec recall

avg of precision & recall avg in denom harmonic mean

Confusion matrix metrics ✓ (5 total)

Another model fit metric: AUC of ROC
$\hat{f}(x)$ quality, more than confusion matrix,

Idea: How well does $\hat{f}(x)$ distinguish $Y=0$ vs $Y=1$? 2 extreme cases

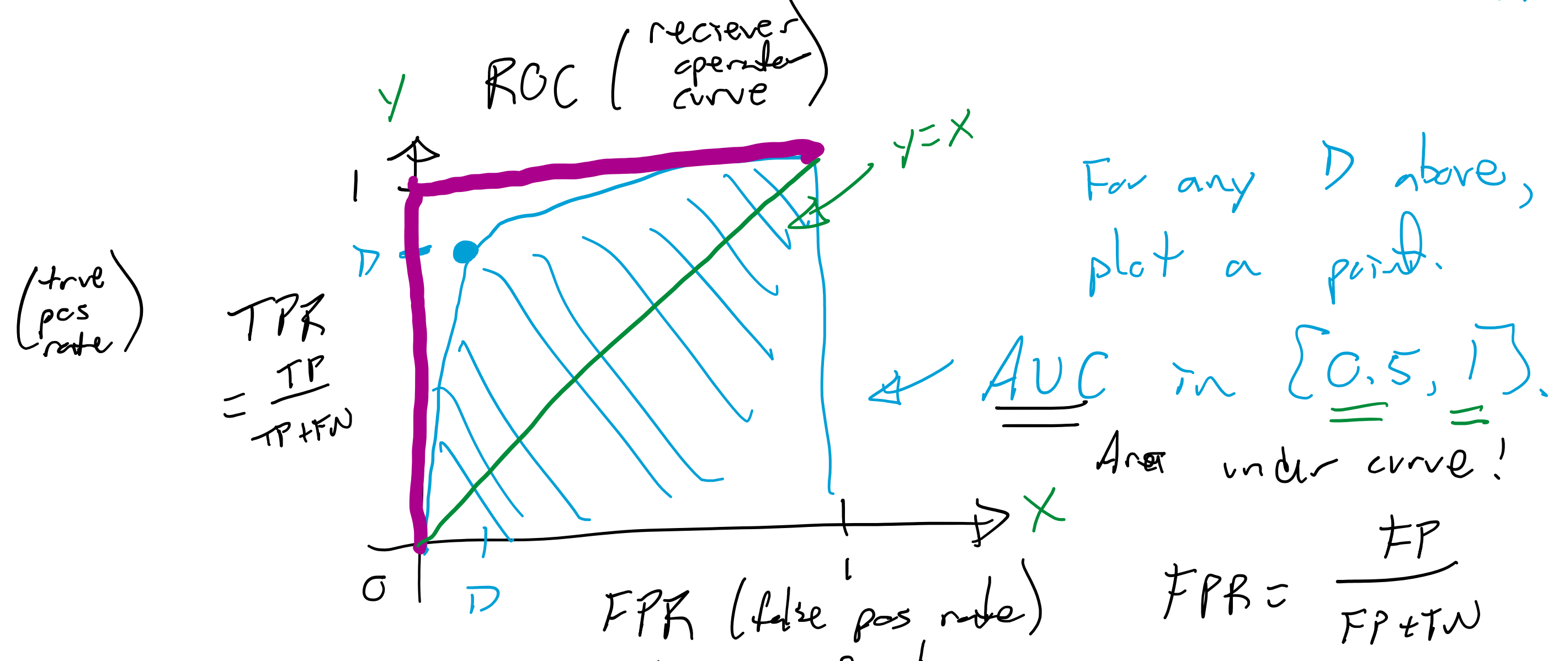


As D decreases
① FPR increases
② TPR increases

Hope $\hat{f}$ has very separate distributions

As D increases
① FPR decreases
② TPR decreases

ROC (reciever operator curve)



(true pos rate) TPR $= \frac{TP}{TP+FN}$

FPR (false pos rate) = 1 − specificity

For any D above, plot a point.

AUC in [0.5, 1]. Area under curve!

$FPR = \frac{FP}{FP+TN}$

Note:
For $C > 2$ classes, do 1 vs rest comparisons & avg all metrics M1–M5 & AUC.

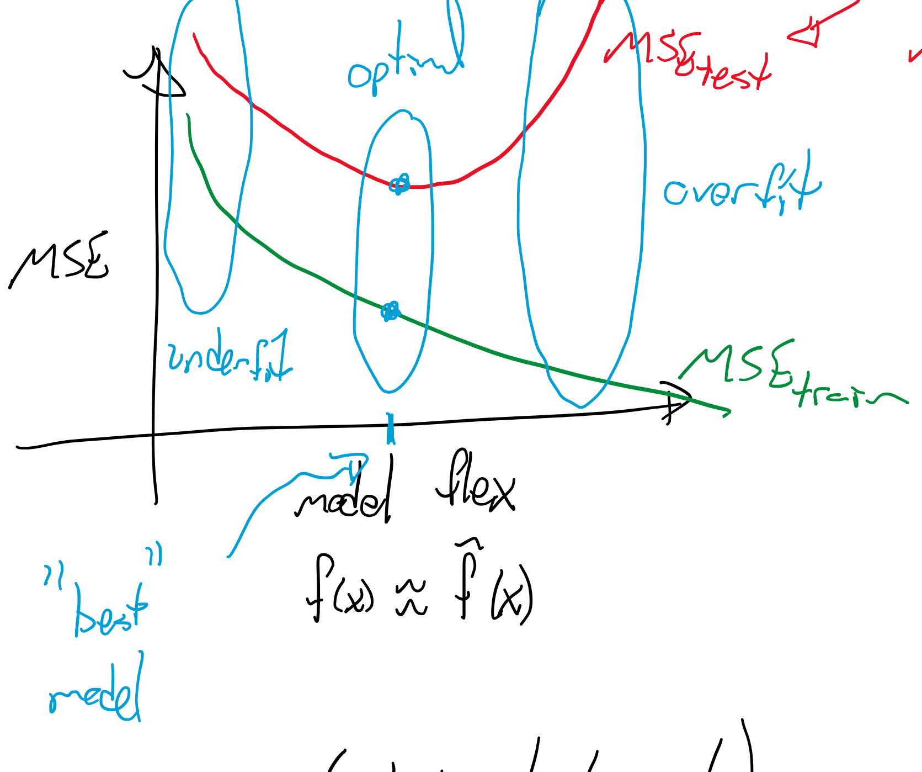
Ch5 Resampling methods

Issue: In practice you don't have a test dataset.
Reason: Train best possible $\hat{f}$ & use all data for train set.

Issues: ① Choose model flexibility / best model — Cross validation ☆
② CI & HTs if don't have SE formulas — Bootstrapping Ch3 ✓
Solve by resampling train dataset

Cross validation Regression: $Y = f(x) + \varepsilon$, $Y \in (-\infty, \infty)$

Ch3



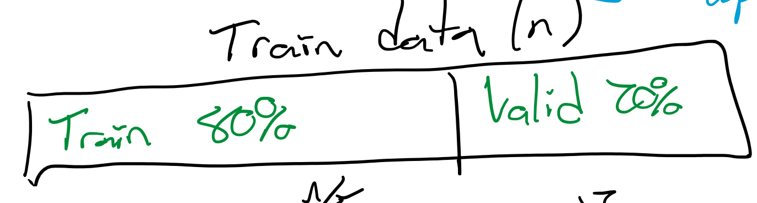
Approximate MSE_{test}

$f(x) \approx \hat{f}(x)$

3 ways to approximate MSE_{test} (w/out test set)

① Validation set.

☆



final model keeps all n to train.

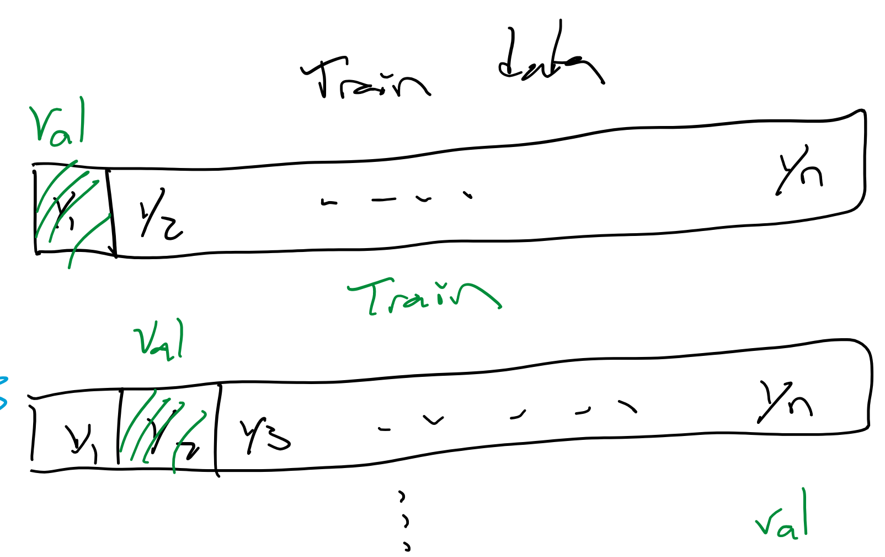
$$MSE_{val} = \frac{1}{n/5} \sum_{i=1}^{n/5} (y_i - \hat{y}_i)^2 \approx MSE_{test}$$

Issue: Validation set is sensitive to random splits

② Repeat validation — many valid. set

ⓐ LOOCV (leave one out cross validation)

n-fold CV



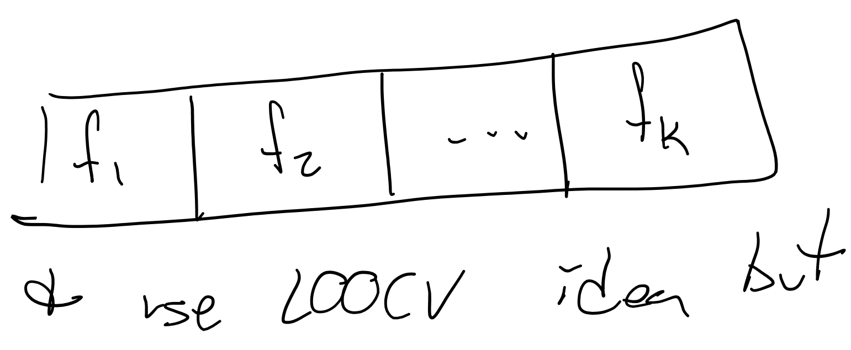
n-models + n-val sets

$$MSE_{LOOCV} = \frac{1}{n} \sum_{i=1}^{n} MSE_i$$

$$MSE_i = (y_i - \hat{y}_i)^2$$

Issue: Inefficient & y_i outliers heavily skew MSE_{LOOCV}

ⓑ k-fold CV



k-subsets of data ← fixed entire process
k = 10 standard

& use LOOCV idea but k times,

$$MSE_k = \frac{1}{k} \sum_{i=1}^{k} MSE_i, \quad MSE_i = \frac{1}{n/k} \sum_{i=1}^{n/k} (y_i - \hat{y}_i)^2$$